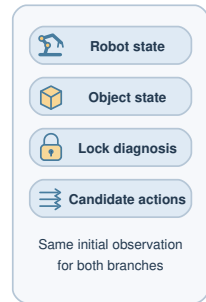
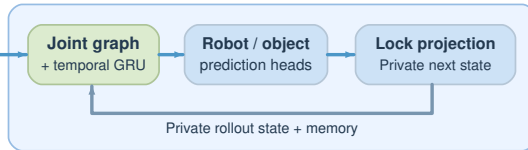


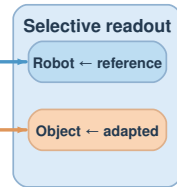
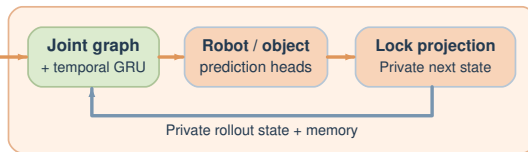
OBSERVATIONS & ACTIONS



REFERENCE BRANCH



ADAPTED BRANCH



Readout is not fed back



TRAINING

Autonomous rollouts · normalized state loss
Freeze robot parameters · update object modules

OBJECT ADAPTATION

Detached robot code + object state → object head
Contact geometry → learned residual → object increment

Blue: preserved robot forecast

Orange: adapted object response

Bottom panels: object adaptation